

0.1 Solution

Let

$$P(x) = x^7 + c_6x^6 + c_5x^5 + c_4x^4 + c_3x^3 + c_2x^2 + c_1x + c_0$$

be a polynomial satisfying the conditions of a *good* polynomial. All coefficients are integers, P is monic, every coefficient is odd and its absolute value is $< 2025^2$, and for every coefficient the number $|c_i|$ is a root of P .

Denote

$$r_i = |c_i| \ (i = 0, \dots, 6), \quad t = r_0 = |c_0|.$$

All r_i are positive odd integers with $1 \leq r_i < 2025^2$. Because $P(r_i) = 0$ and the leading coefficient is 1, the Rational Root Theorem gives $r_i \mid c_0$. Hence $r_i \mid r_0 = t$; in particular t is the largest among the r_i .

0.1.1 1. Factorisation by $x - t$

Since $t = r_0$ is a root, we can factor

$$P(x) = (x - t)Q(x),$$

where $Q(x) = x^6 + a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$ is monic with integer coefficients. Expanding the product yields

$$\begin{aligned} c_6 &= a_5 - t, \\ c_5 &= a_4 - ta_5, \\ c_4 &= a_3 - ta_4, \\ c_3 &= a_2 - ta_3, \\ c_2 &= a_1 - ta_2, \\ c_1 &= a_0 - ta_1, \\ c_0 &= -ta_0. \end{aligned}$$

The constant term gives $|c_0| = t = |-ta_0|$, so $|a_0| = 1$. Thus

$$a_0 = \varepsilon_0, \quad \varepsilon_0 = \pm 1.$$

0.1.2 2. The possible values of r_i

For any index i with $r_i \neq t$ we have $P(r_i) = 0$ and $r_i \neq t$, hence

$$0 = P(r_i) = (r_i - t)Q(r_i) \implies Q(r_i) = 0.$$

Q is monic with integer coefficients, so by the Rational Root Theorem every integer root of Q must divide $a_0 = \pm 1$. Consequently every such r_i must be ± 1 ; being positive it is 1. Therefore

$$r_i \in \{1, t\} \quad \text{for all } i.$$

0.1.3 3. The case $t = 1$

If $t = 1$ then all $r_i = 1$; i.e. every coefficient c_i is either $+1$ or -1 . Condition $P(1) = 0$ (which holds because $r_i = 1$ is a root) gives

$$1 + \sum_{i=0}^6 c_i = 0.$$

Writing p for the number of indices with $c_i = +1$ (the rest are -1), we obtain

$$1 + (p - (7 - p)) = 2p - 6 = 0 \implies p = 3.$$

Thus exactly three of the seven coefficients are $+1$ and the other four are -1 . The leading coefficient is already $+1$, so we must choose which three of the positions $\{0, \dots, 6\}$ receive $+1$. The number of such choices is

$$\binom{7}{3} = 35.$$

All these polynomials satisfy the oddness and size restrictions because $1 < 2025^2$. Hence **there are 35 good polynomials with $t = 1$.**

0.1.4 4. The case $t > 1$

Now suppose $t > 1$ (so t is odd, at least 3). From (3) each coefficient has absolute value either 1 or t . If all $r_i = t$ (i.e. every coefficient has absolute value t), then $P(t) = 0$ forces

$$t^7 + (t\varepsilon_6)t^6 + \dots + t\varepsilon_0 = t^0? \text{ (see below).}$$

A short analysis shows that this is impossible for $t > 1$ (the leading term $t^7(1 + \varepsilon_6)$ cannot be cancelled by the remaining terms because t^5 dominates). Hence at least one coefficient has absolute value 1. For such a coefficient we have $P(1) = 0$, and therefore $Q(1) = 0$ (since $1 \neq t$).

4.1 Recurrence from the coefficients

Write each c_i as

$$c_i = \eta_i r_i, \quad \eta_i = \pm 1, \quad r_i \in \{1, t\}.$$

From (1) we obtain the relations

$$\begin{aligned}
a_5 &= t + \eta_6 r_6, \\
a_4 &= ta_5 + \eta_5 r_5, \\
a_3 &= ta_4 + \eta_4 r_4, \\
a_2 &= ta_3 + \eta_3 r_3, \\
a_1 &= ta_2 + \eta_2 r_2, \\
a_0 &= ta_1 + \eta_1 r_1.
\end{aligned}$$

4.2 Determining a_1 and r_1 Because $a_0 = \pm 1$ and $t > 1$, equation (4f) forces $r_1 = 1$. Indeed, if $r_1 = t$ then $a_0 = t(a_1 + \eta_1)$ is a multiple of $t \geq 3$, contradicting $a_0 = \pm 1$. Hence $r_1 = 1$ and

$$a_0 = ta_1 + \eta_1.$$

Since $a_0 = \pm 1$ and $t \geq 3$, the only possibility is $a_1 = 0$ and $\eta_1 = a_0$. Thus

$$a_1 = 0, \quad r_1 = 1, \quad \eta_1 = a_0.$$

4.3 Determining the remaining indices Now (4e) becomes $0 = ta_2 + \eta_2 r_2$. Hence r_2 must be t (otherwise the right-hand side would be a multiple of t plus η_2 , which cannot vanish with a_2 integer). Moreover

$$a_2 = -\eta_2.$$

Insert (7) into (4d):

$$-\eta_2 = ta_3 + \eta_3 r_3.$$

If $r_3 = t$ the left side is $-\eta_2$ and the right side is a multiple of t plus $\eta_3 t$, forcing $-\eta_2 \equiv 0 \pmod{t}$, impossible because $t \geq 3$. Therefore $r_3 = 1$ and the equation becomes

$$-\eta_2 = ta_3 + \eta_3.$$

Since $t \geq 3$, the only way the right side can be ± 1 is that it is actually 0; thus

$$a_3 = 0, \quad \eta_3 = -\eta_2, \quad r_3 = 1.$$

Repeating the argument, (4c) gives

$$0 = ta_4 + \eta_4 r_4.$$

As before, r_4 cannot be 1 (otherwise ta_4 would have to be an integer not divisible by t), so $r_4 = t$ and

$$a_4 = -\eta_4.$$

Next, (4b):

$$-\eta_4 = ta_5 + \eta_5 r_5.$$

If $r_5 = t$ we would need $-\eta_4 \equiv 0 \pmod{t}$, impossible. Hence $r_5 = 1$ and

$$-\eta_4 = ta_5 + \eta_5.$$

Again this forces the right side to be 0, so

$$a_5 = 0, \quad \eta_5 = -\eta_4, \quad r_5 = 1.$$

Finally, (4a) yields

$$0 = t + \eta_6 r_6.$$

Thus r_6 must be t and $\eta_6 = -1$. Consequently

$$r_6 = t, \quad \eta_6 = -1.$$

4.4 The sign condition and counting Collecting the results we have

$$\begin{aligned} r_6 = t, \eta_6 = -1; & & r_5 = 1, \eta_5 = -\eta_4; \\ r_4 = t, \eta_4 = \pm 1; & & r_3 = 1, \eta_3 = -\eta_2; \\ r_2 = t, \eta_2 & & = \pm 1; r_1 = 1, \eta_1 & & = a_0; \\ a_5 = 0, a_4 & & = -\eta_4, a_3 = 0, a_2 = -\eta_2, a_1 = 0, a_0 & & = \pm 1. \end{aligned}$$

Equation (5) gives $a_0 = ta_1 + \eta_1 = \eta_1$, but also from the recursion we obtain

$$a_0 = ta_1 + \eta_1 = 0 + \eta_1 = \eta_1,$$

while using (6)–(11) we can also express a_0 in terms of η_2, η_4 :

$$a_0 = \eta_1 = -\eta_3 = \eta_2.$$

But from (10) we have $\eta_5 = -\eta_4$ and (8) gives $\eta_3 = -\eta_2$; these do not directly give a_0 . A more systematic evaluation of the recurrence from the bottom shows

$$a_0 = \eta_2 + \eta_4 - 1.$$

(Indeed, substituting all relations into (4f) yields this identity.) Since $a_0 = \pm 1$, we need

$$\eta_2 + \eta_4 - 1 = \pm 1.$$

Both η_2, η_4 are ± 1 . The possibilities are

η_2	η_4	$\eta_2 + \eta_4 - 1$
1	1	1
1	-1	-1
-1	1	-1
-1	-1	-3

Only the first three rows give $a_0 = \pm 1$. Hence exactly **three** choices of the signs (η_2, η_4) are admissible for each fixed $t > 1$.

For each such choice we obtain a concrete polynomial, and all coefficients are odd with absolute values either 1 or t . The condition $P(t) = 0$ holds automatically because t is a root, and $P(1) = 0$ follows from the construction (it is equivalent to $Q(1) = 0$, which is satisfied).

Thus for every odd integer t with $3 \leq t < 2025^2$ there are **exactly three good polynomials**.

4.5 Counting the polynomials for $t > 1$ We need the number of odd integers t in the interval $[3, 2025^2 - 1]$. Since 2025 is odd, 2025^2 is odd, so the largest odd less than 2025^2 is $2025^2 - 2 = 4\,100\,623$. The number of odd integers from 1 to 4 100 623 inclusive is

$$\frac{4\,100\,623 + 1}{2} = 2\,050\,312.$$

Removing $t = 1$ leaves

$$N = 2\,050\,312 - 1 = 2\,050\,311$$

odd values $t \geq 3$ with $t < 2025^2$.

Each such t contributes 3 polynomials, so the number for $t > 1$ is

$$3 \cdot N = 3 \cdot 2\,050\,311 = 6\,150\,933.$$

0.1.5 5. Total number of good polynomials

Adding the 35 polynomials from the case $t = 1$ we obtain

$$35 + 6\,150\,933 = 6\,150\,968.$$

6 150 968
